

Simultaneous State and Parameter Estimation of Aircraft Pitch Dynamics with Extended Kalman Filter and Kalman Smoothing

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1 Summary

One of the most difficult aspects of predicting aircraft dynamic behavior is obtaining accurate values for the aerodynamic coefficients. Though some analytical tools and simulations are available, often the simplest and most accurate method is through parameter estimation based on the actual flight data. For real world systems, the true aircraft dynamics are never perfectly known and instead are only reported through sensor measurements. Therefore, in order to learn the aerodynamic parameters, both the parameters and the aircraft's dynamics must be simultaneously estimated. Through an extended Kalman filter (EKF) the full state was established to estimate both the states describing the aircraft's dynamics and the states describing its parameters. The EKF performed estimation on the full state through a short maneuver. After the maneuver was complete, the states were passed into a Kalman smoother (KS) in an attempt to enhance the estimates generated by the EKF. Experimentation was performed with regard to the effect that the number of unknown parameters had on estimation performance. In addition, the ability of the KS to improve both estimates of the dynamics states and parameters was explored. The KS decreased the estimate error by 59% for the dynamics states but made insignificant difference to the parameter estimates. The parameter estimates themselves decreased in accuracy as the number of unknown parameters increased but many were still capable of convergence on their true value.

2 Goal of Project

The goal of this project was to develop an aircraft model and EKF capable of estimating both states and parameters of an aircraft and its dynamics. The eventual goal of a system like this is to accurately estimate the parameters of a custom aircraft. Experimentation with a KS's ability to estimate parameters and the limitation of unknowns in a dynamic system were also explored.

3 Technical Approach

3.1 Dynamic Model

There are a variety of aircraft dynamic models each using different guiding assumptions. One relatively simple model, the 2D dynamics of a large aircraft in the pitching direction, was chosen. The full derivation is available in "Introduction to Aircraft Stability and Control Course Notes for M&AE 5070", page 81 [2]. Here we use the final equations, a relationship between vertical and

horizontal velocity in the body frame and pitch in the global frame, linearized about horizontal velocity u_0 . This transfer function is shown in equation 1.

$$A = \begin{bmatrix} X_u & X_w & 0 & -g_0 \cos \Theta_0 \\ \frac{Z_u}{1-Z_{\dot{w}}} & \frac{Z_s}{1-Z_{\dot{w}}} & \frac{u_0+Z_q}{1-Z_{\dot{w}}} & \frac{-g_0 \cos \Theta_0}{1-Z_{\dot{w}}} \\ M_u + \frac{M_{\dot{w}}Z_u}{1-Z_{\dot{w}}} & M_w + \frac{M_{\dot{w}}Z_w}{1-Z_{\dot{w}}} & M_q + \frac{(u_0+Z_q)M_{\dot{w}}}{1-Z_{\dot{w}}} & \frac{-M_{\dot{w}}g_0 \sin \Theta_0}{1-Z_{\dot{w}}} \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (1)$$

The state acting on equation 1 is $[u \ w \ q \ \theta]^T$ where u is the body frame horizontal velocity difference from u_0 , w is the body frame vertical velocity, θ is the angle the aircraft's horizontal axis makes with the velocity vector the frame is linearized about and $q = \dot{\theta}$. These components are shown graphically in figure 1 where $u_0 = V_0$. In order to differentiate between quantities relating to dynamics states and parameter states, the subscript $()_s$ will be attached to dynamics states, $()_p$ will be attached to parameter states, and no subscript will be attached to full states which are a combination of the other two.

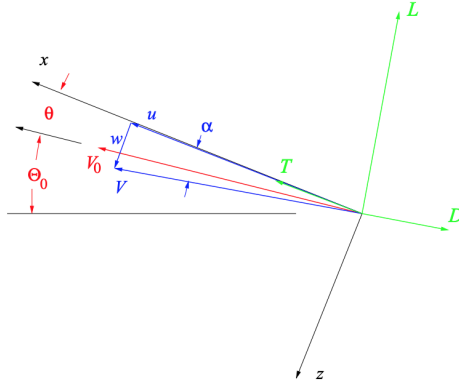


Figure 1: Inertial Frame Diagram

The measurements for this system were simulated as if they came from an inertial measurement unit (IMU) and global positioning system (GPS). The IMU could supply measurements of the angle θ as well as the rate q . The GPS was capable of supplying position measurements. Since there is no position in the current state, the state had to be augmented to include the position of the aircraft. Augmentation of the state required augmentation of the dynamics function by equation 2.

$$\begin{bmatrix} \dot{x}_i \\ \dot{z}_i \end{bmatrix} = \begin{bmatrix} (u + u_0) * \cos(\theta + \Theta_0) + w * \sin(\theta * \Theta_0) \\ -(u + u_0) * \sin(\theta + \Theta_0) + w * \cos(\theta * \Theta_0); \end{bmatrix} \quad (2)$$

where x_i and z_i are the axes of the inertial frame, aligned with x and z when $\Theta_0 = 0$ and $\theta = 0$. Since equation 2 is non-linear and coupled, the full dynamics function becomes equation 3.

$$\begin{bmatrix} \dot{x}_i \\ \dot{z}_i \\ \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \mathbf{f}_s(\mathbf{x}_s) = \begin{bmatrix} (u + u_0) * \cos(\theta + \Theta_0) + w * \sin(\theta * \Theta_0) \\ -(u + u_0) * \sin(\theta + \Theta_0) + w * \cos(\theta * \Theta_0) X_u u + X_w w + 0 + -g_0 \cos \Theta_0 \theta \\ \frac{Z_u}{1-Z_{\dot{w}}} u + \frac{Z_s}{1-Z_{\dot{w}}} w + \frac{u_0+Z_q}{1-Z_{\dot{w}}} q + \frac{-g_0 \cos \Theta_0}{1-Z_{\dot{w}}} \theta \\ M_u + \frac{M_{\dot{w}} Z_u}{1-Z_{\dot{w}}} u + M_w + \frac{M_{\dot{w}} Z_w}{1-Z_{\dot{w}}} w + M_q + \frac{(u_0+Z_q) M_{\dot{w}}}{1-Z_{\dot{w}}} q + \frac{-M_{\dot{w}} g_0 \sin \Theta_0}{1-Z_{\dot{w}}} \theta \\ 0 + 0 + q + 0 \end{bmatrix} \quad (3)$$

with full state $\mathbf{x}_s = [x_i \ z_i \ u \ w \ q \ \theta]^T$.

The parameters of the aircraft dynamics are X_u , X_w , Z_u , Z_w , Z_{wd} , Z_q , M_u , M_w , M_{wd} , and M_q . These parameters are all combinations of more physically important aircraft parameters (C_L , C_D , etc.). However, one of the benefits of this type of estimation process is that the physical significance of a parameter is not relevant to finding its value so we are fine using only the parameters explicitly expressed in the dynamics equations.

The chosen flight condition was a passive maneuver in which the aircraft began in a slightly nose up position and went through minorly dampened oscillations as it flew. The true states of the maneuver are shown in figure 2.

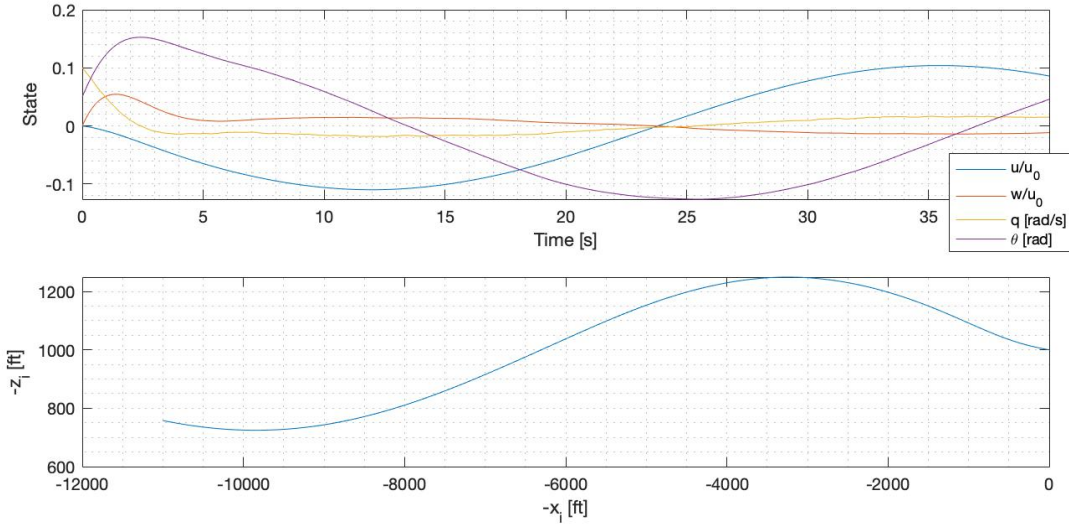


Figure 2: True State Maneuver. Initial condition $\mathbf{x}_{0s} = [0 \ -1000 \ 0 \ 0 \ 0.1 \ 0.05]^T$. Important note is that the aircraft is traveling right to from left in order to align with Figure 1.

3.2 Extended Kalman Filter

The EKF was chosen for this project due to its simplicity and ability to deal with non-linear systems. The general equations guiding the EKF are shown in equation 4.

$$\begin{aligned}
\hat{\mathbf{x}}_{k+1|k} &= f(\hat{\mathbf{x}}_{k|k}) \\
P_{k+1|k} &= F_k P_{k|k} F_k^T + G_k Q_k G_k^T \\
K_{k+1|k} &= P_{k+1|k} H_{k+1}^T (H_{k+1} P_{k+1|k} H_{k+1}^T + R_{k+1})^{-1} \\
\hat{\mathbf{x}}_{k+1|k+1} &= \hat{\mathbf{x}}_{k+1|k} + K_{k+1} (\mathbf{z}_{k+1} - h(\hat{\mathbf{x}}_{k+1|k})) \\
P_{k+1|k+1} &= (I - K_{k+1} H_{k+1}) P_{k+1|k} (I - K_{k+1} H_{k+1})^T + K_{k+1} R_{k+1} K_{k+1}^T
\end{aligned} \tag{4}$$

We assume that there are measurements of position, angle, and pitch rate. In addition, there are aerodynamic disturbances affecting the velocity and pitch rate of the aircraft. These measurements and disturbances translate to the matrices H_s and G_s .

$$H_s = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad G_s = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \tag{5}$$

The Q_s and R matrices representing process and measurement noise are shown in equation 6,

$$Q_s = \begin{bmatrix} 0.01^2 & 0 & 0 \\ 0 & 0.01^2 & 0 \\ 0 & 0 & 0.001^2 \end{bmatrix} \times \Delta t \quad R = \begin{bmatrix} 3^2 & 0 & 0 & 0 \\ 0 & 3^2 & 0 & 0 \\ 0 & 0 & 0.1^2 & 0 \\ 0 & 0 & 0 & 0.1^2 \end{bmatrix} \tag{6}$$

where Δt represents the timestep of the EKF updates. For the results in this report, $\Delta t = 0.1s$. The dynamics of the true system were generated using `ode45` with equation 3.

3.3 Augmenting State with Parameters

In order to estimate the parameters, the parameters first had to be added to the state vector. For implementation, MATLAB's symbolic toolbox was leveraged to generate the jacobian.

The full state is defined as $\mathbf{x} = [\mathbf{x}_s^T \mathbf{x}_p^T]^T$ where

$$\mathbf{x}_p = [X_u \quad X_w \quad Z_u \quad Z_w \quad Z_{wd} \quad Z_q \quad M_u \quad M_w \quad M_{wd} \quad M_q]^T \tag{7}$$

The full time update function becomes,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \begin{bmatrix} \mathbf{f}_s(\mathbf{x}_s) \\ \mathbf{0} \end{bmatrix} \tag{8}$$

where $\mathbf{0}$ is a 10 element column vector of zeros. The jacobian is defined with equation $J = \frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}$.

Since the true system is assumed to have static parameters, the process noise acting on the parameters, Q_p , is empty. Though these parameters are valuable for tuning the system, they are left as a point of exploration for a future report. Therefore, $G_p = [0 \quad 0 \quad 0]^T$. The initial parameter covariance matrix, P_{0p} , was given covariances equal to that parameter's value multiplied by standard deviation $\sigma_p = 0.5$, all squared. This allowed the error in each estimate to be a similar

order of magnitude as the true parameter. It is assumed that analytical calculations would provide an initial estimate of the flight parameters to this level of accuracy.

Similarly, the initial states were expressed as each parameter times $(1 + \sigma_p \times \mathbf{randn})$ in order to control their error distribution.

The full states were augmented as

$$H = H_s \quad G = G_s \quad P = \begin{bmatrix} P_s & 0 \\ 0 & P_p \end{bmatrix} \quad (9)$$

where P_s is the covariance of the dynamic states.

3.4 State and Covariance Update Functions

The EKF utilized two specific functions: one that generated state predictions and the other that generated the transition matrix, F , at every time step. The state prediction merely implemented equation 8 with `ode45` for a time step of Δt . The transition matrix was approximated using the jacobian instead of calculated with an infinite sum or inverse Laplace transform. This approximation inherently made the assumption that the time steps were small compared to the dynamics. F was calculated with equation 10.

$$F = I + \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \Delta t \quad (10)$$

where I is the identity matrix.

3.5 Kalman Smoothing

Kalman smoothing was implemented to provide predictions for $\mathbf{x}_k$ that take advantage of both the previous estimated states, $\mathbf{x}_{0:k-1}$, and those after, $\mathbf{x}_{k+1:N}$. These new estimates were expected to be smoother than the original estimates from the EKF and potentially more accurate. One hope was that the lowest time step KS estimate, the last of its estimates, might provide a more accurate estimate of the parameters than the last EKF estimate, its most accurate.

The KS uses the equations,

$$\begin{aligned} K_k^* &= P_{k|k} F_k^T P_{k+1|k}^{-1} \\ \hat{\mathbf{x}}_{k|N}^* &= \hat{\mathbf{x}}_{k|k} + K_k^* \left[\hat{\mathbf{x}}_{k+1|N}^* - \hat{\mathbf{x}}_{k+1|k} \right] \\ P_{k|N}^* &= P_{k|k} + K_k^* \left[P_{k+1|N}^* - P_{k+1|k} \right] (K_k^*)^T \end{aligned} \quad (11)$$

where $P_{N|N}^*$ and $\hat{\mathbf{x}}_{N|N}^*$ are initiated as the last covariance matrix and state estimate of the EKF and time is stepped backwards. F is the transition matrix for the state $\hat{\mathbf{x}}_{k+1|N}^*$.

4 Results

4.1 No Unknown Parameters

The initial task was to verify the EKF's ability to estimate the dynamic states with fully known parameters. Observe from figure 3 the state estimate error for both the forward EKF and KS.

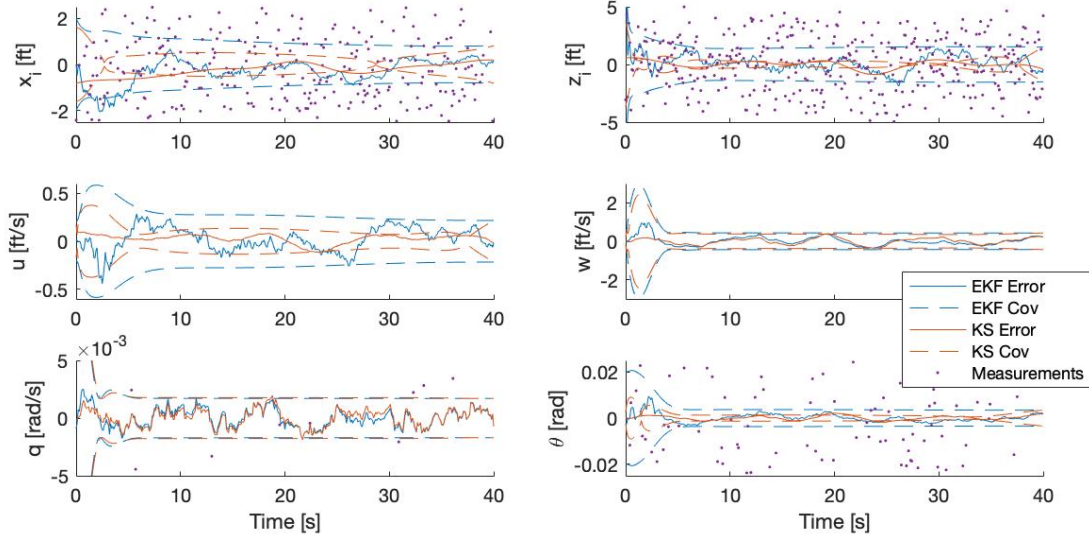


Figure 3: State estimate error for 0 unknown parameters. The KS provides smoother and more accurate overall estimates.

Measured states are estimated to significantly below the measurement noise covariances, a strong sanity check that the filter is working. The improvement provided by the KS was present for every state but varied significantly in magnitude. Since the mean of each estimate was near zero, the standard deviation of the error was the metric used for measuring performance. For the x_i , y_i , u , and w , the standard deviation decreased by about 50% while for q and θ the decrease was closer to 15%. Full statistics are presented in table 1.

4.2 1 Unknown Parameter

The same flight maneuver was repeated with one unknown parameter, X_u . Observe in figure 4 that the estimates of each dynamics state behaved very similarly to how they did with 0 unknown parameters, though the average standard deviation increased by 52%. This is because without a known parameter, the predicted dynamics were also less well known.

Figure 5 demonstrates very interesting results. With only 1 unknown parameter, the EKF was capable of generating a very accurate estimate after about 17 seconds. The smoother maintained the near perfect estimate from 40 to 5 seconds, after which it diverged to a worse estimate. The divergence over the last 5 seconds is small and likely caused by the significantly larger range of estimates present at the beginning of the maneuver. This result suggests that the KS's utilization of the previous estimates does not effect parameters, a state with zero natural change, as much as the dynamic states.

4.3 4 Unknown Parameters

As an intermediate step between 1 unknown parameter and all 10 unknown parameters, the same process was repeated for the first 4 parameters as unknowns. Here, the standard deviation of the

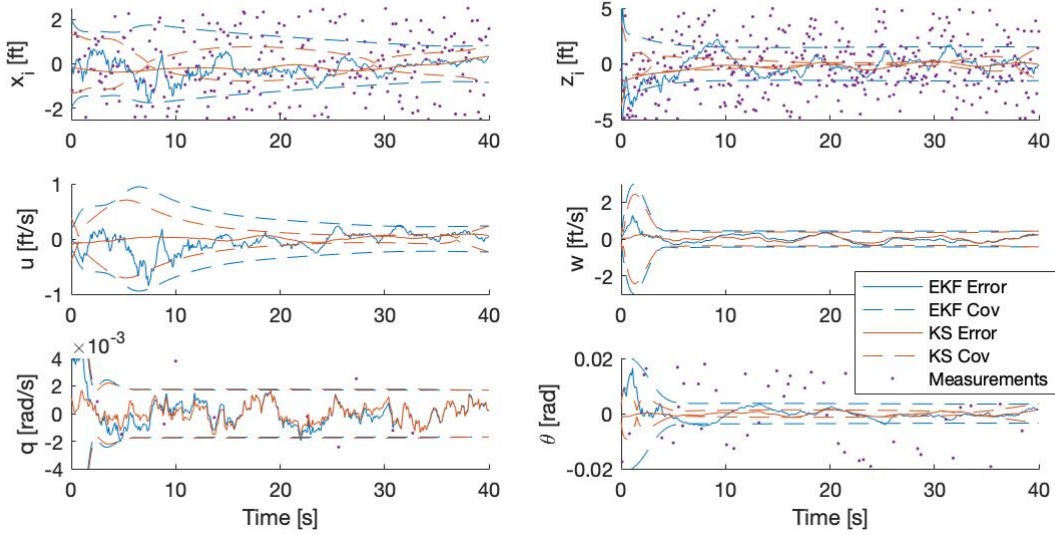


Figure 4: State estimate error with 1 unknown parameter.

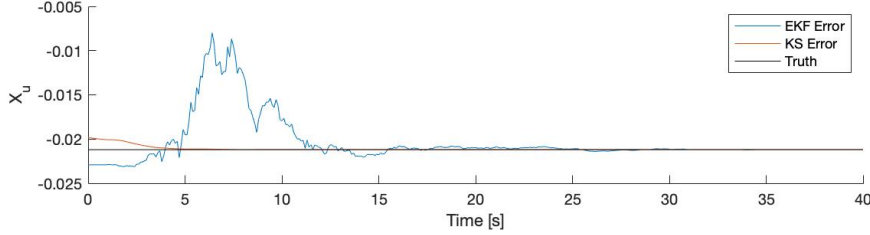


Figure 5: Parameter estimate error of 1 unknown parameter. The error in the final state of the EKF (right most point) is smaller than in the final state of the KS (left most point).

dynamics states estimate errors for the EKF barely changed from those of the 1 unknown parameter case, with an average increase of only 0.27%.

Figure 7 demonstrates that with more unknown parameters, the error in each estimate increases. The percent error in X_u increased by 65% for the 4 unknown parameter system over the single unknown parameter. However, each final estimate was still an improvement over the initial estimate, especially for X_w , Z_u , and Z_w where the error decreased by about 80% for each. The Kalman smoother's estimates performed inconsistently as before. Though the estimate of X_w improved slightly through the KS, the estimates of the other three were worse, further disproving the enhanced estimate possibility. Full results of parameter estimate error are displayed in table 2.

4.4 10 Unknown Parameters

Though the capabilities of the EKF to estimate parameters degraded with the number of unknown parameters, it was of interest to observe the performance with all 10 parameters treated as unknowns. The average standard deviation of the dynamics states increased by a factor of 5.3 from the 4 to 10 unknown parameter configurations, primarily due to increases in the states w , q , and

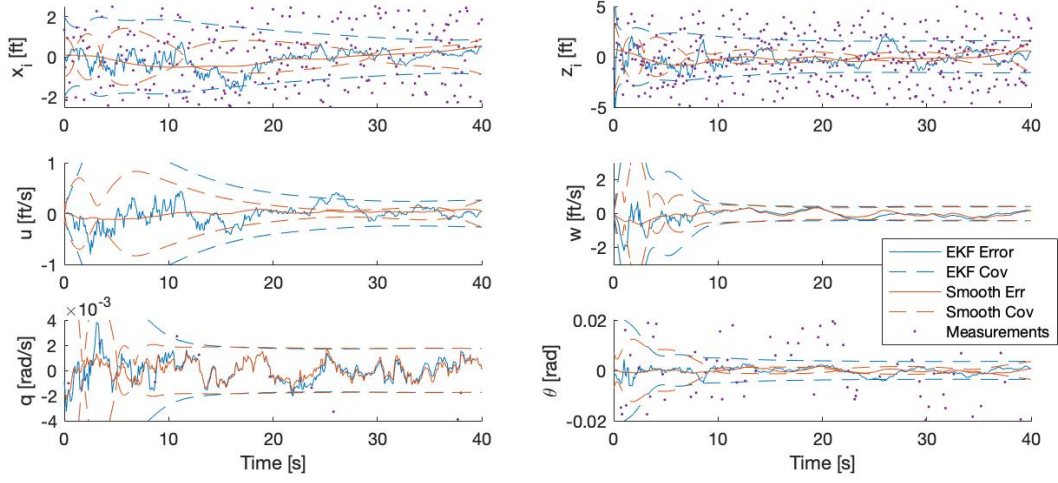


Figure 6: State estimate error with 4 unknown parameters.

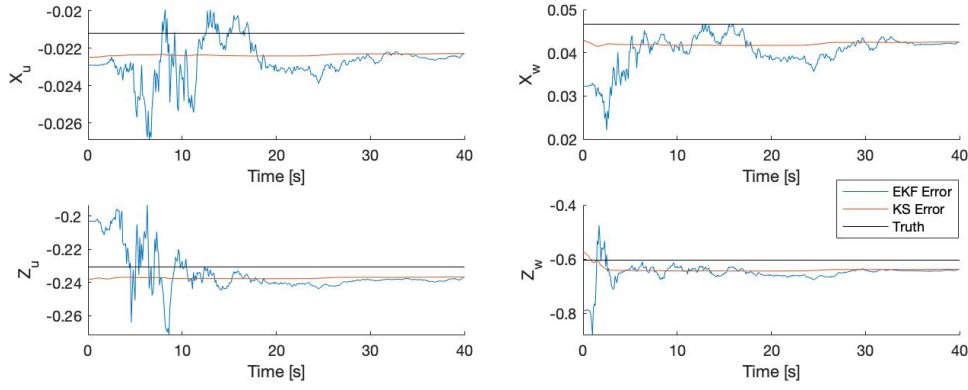


Figure 7: Parameter estimate error of 4 unknown parameters.

θ . It is hard to understand why exactly these states suffered more than the others from poor parameter knowledge, but it is likely due to the relationship these states have in the dynamics. Also note here that the mean of these three state estimate errors is no longer close to 0, suggesting that the standard deviation metric does not capture all of the error present in these estimates.

Observer in figure 9 that the estimates of each parameter vary significantly from each other. A few states, M_u , M_w , and M_q , converge well on their true values while the rest show very little improvement or even converge on an incorrect estimate. The false convergence suggests some sort of local minimum in the estimation algorithm where the incorrect parameter estimates reinforce other incorrect parameter estimates. This would be a good situation to test out the effect of process noise Q_p in future work. As before, we observe that the final KS estimate behaves poorly, only decreasing the final estimate error in 3 of the 10 cases.

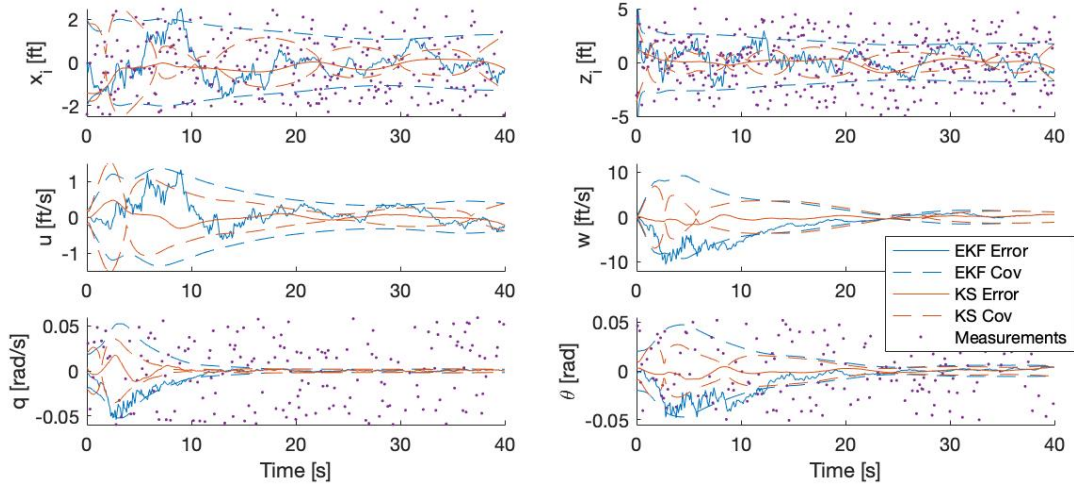


Figure 8: The state estimate error with 10 unknown parameters. States w , q , and θ express significantly more error than in other situations.

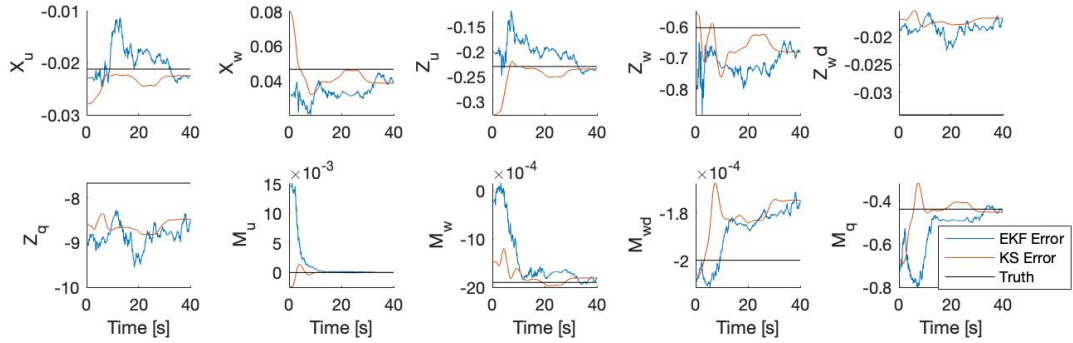


Figure 9: Parameter estimate error of 10 unknown parameters.

4.5 Cumulative Tables

The effects of increasing unknown parameters and the capabilities of the KS are summarized in tables 1 and 2 where 1 displays the standard deviation of the dynamics states estimate error and 2 communicates the percent error of the final estimates of the parameters. The KS decreased the standard deviation of dynamics state estimates by an average 59%, demonstrating the value of both future and past knowledge for estimation. In addition, the increasing unknown parameters increased the standard deviation of each state as the result of the poorer knowledge of the true model.

The percent error in the estimates of parameters grew with the number of unknown parameters in all cases but one. This is again due to the propagated effect that each estimate has on the others. The variability in the convergence of each parameter also suggests that the parameter estimates depend significantly on how the model dynamics are projected onto the parameters. It would be worth exploring in the future if different types of maneuvers increase convergence on different

State	Standard Deviation					
	x_i	z_i	u	w	q	θ
0 Unknown EKF	5.31E-01	1.21E+00	1.26E-01	2.39E-01	8.18E-04	1.91E-03
0 Unknown KS	2.69E-01	3.27E-01	4.74E-02	1.66E-01	7.64E-04	9.11E-04
1 Unknown EKF	4.23E-01	3.72E+00	1.82E-01	2.42E-01	1.12E-03	2.82E-03
1 Unknown KS	1.54E-01	3.72E-01	4.30E-02	1.31E-01	6.76E-04	6.71E-04
4 Unknown EKF	4.76E-01	3.25E+00	1.90E-01	3.69E-01	9.51E-04	1.69E-03
4 Unknown KS	2.91E-01	3.41E-01	5.68E-02	1.95E-01	7.23E-04	6.83E-04
10 Unknown EKF	7.76E-01	1.39E+00	3.39E-01	2.84E+00	1.19E-02	1.35E-02
10 Unknown KS	3.28E-01	3.76E-01	1.35E-01	5.04E-01	3.68E-03	2.56E-03

Table 1: Summary of dynamics state estimate Error. All values in the units of the related state.

State	% Error Final Estimate									
	X_u	X_w	Z_u	Z_w	Z_{wd}	Z_q	M_u	M_w	M_{wd}	M_q
1 Unknown EKF	0.1									
1 Unknown KS	6.5									
4 Unknown EKF	5.1	8.8	2.6	5.5						
4 Unknown KS	6.1	7.8	3.3	5.8						
10 Unknown EKF	5.9	17.2	2.1	12.9	51.0	10.5	99.5	4.8	12.6	3.2
10 Unknown KS	31.6	72.3	42.0	9.0	50.6	11.9	124.7	23.7	3.1	54.0

Table 2: Summary of error in estimated parameters.

parameters.

The Kalman smoothing did not increase the accuracy of the parameter estimate. On average, the KS percent error final estimates were 19% worse than those of the EKF. Given these results, the Kalman smoother should not be used for enhanced parameter estimation for this type of system.

5 Lessons Learned

First off, I thoroughly enjoyed pursuing this exploration. I gained significant experience applying the EKF to an unknown system as well as became very familiar with many of its inner workings through debugging. I was also amazed by the results of the KS on the dynamics states. After this full exploration, I feel that I've actually generated more questions than found answers. I'd like to see what the effect of a tuning process noise would be as well as different maneuvers. Eventually, I'd also like to implement a system like this in a custom autopilot system designed for stabilization of a remote control aircraft. I hope that this report has sparked others' curiosity as well.

References

- [1] Grillo, Caterina. "An Extended Kalman Filter-Based Technique for On-Line Identification of Unmanned Aerial System Parameters." *SciELO*, July 2015, www.scielo.br/scielo.php?script=sci_arttext&pid=S2175-91462015000300323e10.
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